Functional Approximation Using Tucker ALS

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1 Problem Setup

We want to approximate a smooth function

$$f(x, y, z) = \frac{1}{1 + \|C[x, y, z]^T\|^2}, \quad (x, y, z) \in [-1, 1]^3,$$

using a Tucker representation built directly from Chebyshev basis matrices.

Let

$$T_x \in \mathbb{R}^{N_x \times (d_x+1)}, \quad T_y \in \mathbb{R}^{N_y \times (d_y+1)}, \quad T_z \in \mathbb{R}^{N_z \times (d_z+1)}$$

be Chebyshev Vandermonde matrices:

$$(T_x)_{i,k} = T_k(x_i), \quad (T_y)_{j,k} = T_k(y_j), \quad (T_z)_{\ell,k} = T_k(z_\ell).$$

We seek the approximation

$$F(x_i, y_j, z_\ell) \approx \sum_{a=1}^{R_x} \sum_{b=1}^{R_y} \sum_{c=1}^{R_z} (T_x A_x)_{i,a} \, (T_y A_y)_{j,b} \, (T_z A_z)_{\ell,c} \, G_{abc}.$$

Equivalently,

$$F \approx (T_x A_x, \ T_y A_y, \ T_z A_z, \ G),$$

a Tucker tensor in value space.

2 ALS Update Structure

The ALS loop alternates three updates:

$$A_x, G \leftarrow \text{argmin}, \quad A_y, G \leftarrow \text{argmin}, \quad A_z, G \leftarrow \text{argmin}.$$

Throughout these steps we use the identity

$$\text{vec}(U_x G_{(1)} (U_z \otimes U_y)^T) = (U_z \otimes U_y \otimes U_x) \, \text{vec}(G).$$

2.1 Notation

Define

$$U_x = T_x A_x, \quad U_y = T_y A_y, \quad U_z = T_z A_z.$$

3 Mode–1 Update

We rewrite F with the x-mode singled out:

$$F_{(1)} \approx U_x \, G_{(1)} (U_z \otimes U_y)^T.$$

Flatten the function values as

$$F_{(1)} \in \mathbb{R}^{N_x \times (N_y N_z)}.$$

Let

$$YZ = U_z \otimes U_y \in \mathbb{R}^{(N_y N_z) \times (R_y R_z)}.$$

We solve the least-squares problem

$$\min_{\tilde{G}_x} \| F_{(1)} - T_x \tilde{G}_x \, YZ^T \|_F.$$

Vectorizing:

$$\operatorname{vec}(F_{(1)}) = (YZ \otimes T_x) \operatorname{vec}(\tilde{G}_x).$$

Solve:

$$\operatorname{vec}(\tilde{G}_x) = (YZ \otimes T_x)^\dagger \operatorname{vec}(F_{(1)}).$$

Reshape:

$$\tilde{G}_x \in \mathbb{R}^{(d_x+1) \times (R_y R_z)}.$$

Apply an SVD:

$$\tilde{G}_x = U_x^c \, S_x \, V_x^T.$$

Truncate:

$$A_x = U_x^c(:, 1 : R_x),$$

$$G_{(1)} = S_x(1 : R_x, 1 : R_x) \, V_x(1 : R_x, :).$$

Reshape $G_{(1)}$ back into a core tensor G via its mode–1 unfolding..

4 Mode–2 Update

Permute F into shape (N_y, N_x, N_z) and flatten:

$$F_{(2)} \in \mathbb{R}^{N_y \times (N_x N_z)}.$$

Define

$$XZ = U_z \otimes U_x.$$

Solve

$$\operatorname{vec}(F_{(2)}) = (XZ \otimes T_y) \operatorname{vec}(\tilde{G}_y).$$

Compute $\tilde{G}_y$, SVD, truncate:

$$A_y = U_y^c(:, 1 : R_y),$$

update G via its mode–2 unfolding.

5 Mode–3 Update

Permute F into (N_z, N_x, N_y):

$$F_{(3)} \in \mathbb{R}^{N_z \times (N_x N_y)}.$$

Define

$$XY = U_y \otimes U_x.$$

Solve

$$\mathrm{vec}(F_{(3)}) = (XY \otimes T_z)\,\mathrm{vec}(\tilde{G}_z).$$

Do SVD, truncate:

$$A_z = U_z^c(:, 1 : R_z),$$

and update G through its mode–3 unfolding.

6 Final Approximation

After convergence, the approximation is

$$\hat{F}(x, y, z) = \sum_{a,b,c} (T_x A_x)_{i,a}\,(T_y A_y)_{j,b}\,(T_z A_z)_{\ell,c}\,G_{abc}.$$

Or equivalently, the Tucker coefficient tensor in Chebyshev space is

$$C_{\text{tucker}} = A_x \times_1 G \times_2 A_y \times_3 A_z,$$

computed explicitly as

$$C_{\text{tucker}}(i, j, k) = \sum_{a,b,c} A_x(i, a) A_y(j, b) A_z(k, c)\,G_{abc}.$$

7 Main Computational Bottleneck

Each ALS step requires solving a linear system of the form

$$\mathrm{vec}(F_{(n)}) = (U_{\text{other}} \otimes T_n)\ \mathrm{vec}(\tilde{G}_n),$$

where

$$U_{\text{other}} \in \mathbb{R}^{(N_i N_j) \times (R_i R_j)}, \qquad T_n \in \mathbb{R}^{N_n \times (d_n + 1)}.$$

The Kronecker product

$$U_{\text{other}} \otimes T_n \in \mathbb{R}^{(N_x N_y N_z) \times ((d_n+1) R_i R_j)}$$

is extremely large. Thus, forming this matrix explicitly and solving

$$\min_{\tilde{G}_n} \left\| \mathrm{vec}(F_{(n)}) - (U_{\text{other}} \otimes T_n)\,\mathrm{vec}(\tilde{G}_n) \right\|_2$$

is the dominant computational cost.

If $N_x = N_y = N_z = N$, then the design matrix has size

$$(N^3) \times \big((d_n + 1)R_iR_j\big),$$

and storing the Kronecker product costs

$$O(N^3(d_n + 1)R_iR_j),$$

Take a typical configuration used in our experiments:

$$N_x = N_y = N_z = 640, \qquad d_n = 63, \qquad R_i = R_j = 10.$$

The design matrix then has

$$N^3 = 640^3 = 26{,}21{,}44{,}000$$

rows, and

$$(d_n + 1)R_iR_j = 64 \times 10 \times 10 = 6{,}400$$

columns.

Thus the design matrix contains

$$26{,}21{,}44{,}000 \times 6{,}400 = 16, 77, 72, 16, 00, 000$$

entries.

8 Extension to Higher Dimensions

Let a d-dimensional function be sampled on a tensor grid, and let

$$T_k \in \mathbb{R}^{N_k \times (m_k+1)}, \qquad k = 1, \ldots, d,$$

denote the Chebyshev Vandermonde matrices. We seek a Tucker representation

$$F \approx (T_1A_1,\ T_2A_2,\ \ldots,\ T_dA_d,\ G),$$

with factor matrices

$$A_k \in \mathbb{R}^{(m_k+1)\times R_k},$$

and core tensor

$$G \in \mathbb{R}^{R_1\times\cdots\times R_d}.$$

ALS Update for Mode n

Define

$$U_k = T_kA_k, \qquad k = 1, \ldots, d.$$

Let $F_{(n)}$ be the mode-n unfolding. Let

$$U_{\neg n} = U_d \otimes \cdots \otimes U_{n+1} \otimes U_{n-1} \otimes \cdots \otimes U_1.$$

Then the mode-n update solves

$$\mathrm{vec}(F_{(n)}) = (U_{\neg n} \otimes T_n)\ \mathrm{vec}(\tilde{G}_n),$$

yielding $\tilde{G}_n \in \mathbb{R}^{(m_n+1)\times(R_1\cdots R_{n-1}R_{n+1}\cdots R_d)}$.

Compute the SVD

$$\tilde{G}_n = U_n^c \Sigma_n V_n^T,$$

truncate

$$A_n = U_n^c(:, 1 : R_n),$$

and update the core via the truncated unfolding

$$G_{(n)} = \Sigma_n(1 : R_n, 1 : R_n)\, V_n(1 : R_n, :).$$

Repeat for $n = 1, \ldots, d$ until convergence.

Final Approximation

$$\hat{F}(i_1, \ldots, i_d) = \sum_{a_1=1}^{R_1} \cdots \sum_{a_d=1}^{R_d} \left(\prod_{k=1}^{d} (T_k A_k)_{i_k, a_k} \right) G_{a_1,\ldots,a_d}.$$